

RNA Structure Before the Experiment

An ensemble view of therapeutic RNA design

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When RNA itself is the therapeutic molecule, constructs are designed, specified, ordered and compared as sequences. When RNA is the therapeutic target, target sites are likewise usually defined by sequence. In either case, the RNA inside a tube or a cell is not just a string of letters. It is a physical object that can begin to fold as it is synthesised, and continues to fold and unfold as it is translated, bound, modified and degraded.

Sequence changes can alter the folding landscape of an RNA even when structure is not the intended target of the design. A codon substitution, for example, may change which regions are exposed or sequestered, the relative populations of alternative conformations, and the overall stability of the molecule. These effects are usually not visible in a construct map, but they can influence how an RNA behaves.

Structural analysis cannot tell us whether a therapeutic concept will work. It can, however, show how a sequence change is expected to affect RNA structure and help turn those effects into experimentally testable hypotheses.

1 A sequence change is also a structural intervention

A silent codon substitution changes no amino acid. It can change the set of base pairs the molecule can form, and the relative weights of the structures in that set. A change intended as neutral at the level of the protein need not be neutral at the level of the RNA fold.

This is not a new observation. Across 154 synonymous variants of a single gene, Kudla and colleagues [1] found that codon bias did not correlate with expression, whereas predicted mRNA folding stability around the ribosome binding site explained more than half of the variation in protein levels.

The evidence is strongest for mRNA. Mauger and colleagues [2] showed that increasing sec-

ondary structure in the coding region raised protein output, and that the effect was primarily associated with a longer functional half-life rather than a higher translation rate. Leppek and colleagues [3], measuring degradation and ribosome load across a large designed panel, found stability and translation could be co-optimised rather than traded off. Zhang and colleagues [4] optimised codon usage and folding energy jointly, and their designed vaccine mRNAs showed longer half-life, higher expression and substantially higher antibody titres in mice.

None of these results says that a more structured mRNA is a better mRNA. What they establish is that sequence design also changes structural properties, whether those changes are considered explicitly or not. A design process that tracks only codon usage and untranslated-region identity therefore leaves part of the molecule's behaviour unexamined.

In other applications, structure can be the design target itself. In our recent work on exoribonuclease-resistant RNAs [5], sequence design was guided by the three-dimensional topology required for resistance to XRN1 rather than by sequence conservation alone. Synthetic sequences with little similarity to known xrRNAs nevertheless reproduced the required ring-like topology and remained functional *in vitro*. This is a more specialised example, but it shows that sequence design can be guided directly by structural principles.

2 The relevant object is a distribution

A standard thermodynamic folding calculation is often summarised by the minimum free energy structure, i.e. the single fold with the lowest predicted free energy. This is a useful representation, but an incomplete one. At finite temperature, an RNA generally populates an ensemble of conformations with probabilities determined by their free energies.

Under an equilibrium thermodynamic model, that distribution is computable. McCaskill's partition function algorithm [6] gives base pair probabilities over the whole ensemble rather than a single fold, and Ding and Lawrence's stochastic sampling [7] draws representative structures from it. Partition function calculations and Boltzmann-weighted stochastic sampling are both implemented in widely used RNA folding packages, including the ViennaRNA Package [8].

The ensemble also tells us something about how much confidence to place in a predicted structure. Mathews [9] showed that base pairs in the minimum free energy structure are more likely to be correct when they also have high pairing probabilities. Those probabilities therefore help identify which parts of a predicted structure are well supported and which are not. Two sequences can have the same minimum free energy structure and still differ substantially in their ensembles. In one case, that structure may dominate the ensemble, while in another, substantial probability may be distributed over competing conformations. A single structure representation would miss that difference, whereas the ensemble makes it visible.

Wayment-Steele and colleagues [10] showed that the average unpaired probability (AUP) of an mRNA is related to its predicted rate of in-line hydrolysis. Designs with lower AUP were predicted to have longer hydrolytic half-lives while retaining properties associated with efficient translation. This also illustrates why an ensemble description can be useful in practice. Rather than asking whether a variant simply "changes the structure", one can ask how much it changes the ensemble, and whether those changes occur in a region relevant

to the question being studied.

3 Accessibility: useful, and not sufficient

For mechanisms that require access to an unpaired region, including antisense oligonucleotides (ASOs), siRNAs, RNA-binding proteins that recognise single-stranded RNA, primers and engineered sensors, it is usually more informative to consider the energetic cost of making a site accessible over the ensemble than whether it happens to be paired in a single predicted structure.

Target accessibility has been used in this way for both siRNA and microRNA target prediction. Tafer and colleagues [11] found that accessibility helped distinguish functional from non-functional siRNAs and improved selection when combined with sequence and asymmetry criteria. Kertesz and colleagues [12] incorporated the energetic cost of opening the target site into microRNA target prediction together with duplex formation energy. In both cases, accessibility contributed useful information, but was not sufficient on its own.

Vickers and colleagues [13] examined identical antisense target sites placed in different structural contexts. Increased target structure reduced antisense activity in cells, but this effect could be partly overcome by using higher-affinity oligonucleotide chemistry. The result is a useful reminder that accessibility matters, but does not act independently of the other properties of the oligonucleotide and target.

Cellular RNA adds another layer of complexity. Thermodynamic models describe a naked RNA at equilibrium, whereas in cells RNA is bound by proteins and remodelled by energy-dependent processes. Rouskin and colleagues [14] found that mRNAs were substantially less structured in living cells than after refolding *in vitro*, and that ATP depletion increased RNA structure. Equilibrium calculations are therefore better viewed as estimates of intrinsic folding behaviour than as direct representations of the structures present in a cell.

4 Folding takes time, and time is part of the phenotype

Equilibrium is an assumption, and for an RNA that is synthesised directionally over seconds it can be a substantial one. RNA can fold co-transcriptionally, so that the 5' end has already begun to form structure before the 3' end is available [15]. As a consequence, thermodynamically favourable structures may not be reached on biologically relevant timescales, while metastable conformations can persist once they have formed.

The effect can be seen directly in regulatory RNAs. Helmling and colleagues [16] combined NMR analysis of individual transcription intermediates with cotranscriptional folding simulations of a riboswitch. The simulations recapitulated the sequence of metastable conformations seen experimentally and provided a mechanistic explanation of how these intermediates contribute to regulation.

This kind of analysis requires models that describe not only which structures are thermodynamically accessible, but also how folding proceeds over time. Our recent work on KinPFN [17] approaches this problem with a machine learning method based on prior-data fitted neural networks. Rather than carrying out the full kinetic simulation for every query, KinPFN learns to approximate the distribution of RNA folding times from simulated training

data and a small number of initial folding-time samples.

5 From ensemble shift to testable hypothesis

In practice, ensemble calculations are most useful when they connect a sequence change to a specific experimental question:

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- ↓ sequence change
 - ↓ ensemble shift
 - ↓ structural consequence
 - ↓ experimental hypothesis
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For a given variant, one can first ask whether the sequence change alters the predicted ensemble, and by how much. If it does, the next question is whether this changes a feature relevant to the mechanism under study. This may be a local region, such as a translation initiation region, target window, protein-binding site or designed switch, but it can also be a more global property of the ensemble. The structural difference becomes experimentally useful when it can be linked to a specific effect that can be measured and, in principle, ruled out.

For such an analysis to be informative, the modelling assumptions need to be stated clearly. This includes the energy model and parameter set, temperature and, where applicable, ionic conditions, the regions that were analysed, and the size of the predicted effect relative to variation among comparable sequences. The report should also make clear which observation would be inconsistent with the proposed interpretation.

6 What this does not do

Structural analysis does not predict therapeutic efficacy. Delivery, innate immune sensing, manufacturing purity and cell type all influence what happens between a predicted RNA structure and an experimental or therapeutic outcome, and none of these factors is represented in a secondary structure model. Nucleotide chemistry is a somewhat different case. Some modified nucleotides can now be treated explicitly, but the available thermodynamic parameters are still far from complete.

Most secondary structure calculations are based on nearest neighbour parameter sets that were originally developed for RNA containing the four standard nucleotides [18]. Parameters have since become available for several modified nucleotides, including N⁶-methyladenosine [19], inosine [20], pseudouridine [21], and N¹-methylpseudouridine [22]. The ViennaRNA Package can incorporate modification-specific energy corrections for a number of such residues [23]. The coverage is nevertheless uneven, and a calculation is only as reliable as the parameterisation available for the modification and structural context being considered.

Experimental measurements can provide additional information, but they do not remove

the modelling step. Structure probing experiments, for example, provide reactivities that can be used to constrain a predicted ensemble, but converting those measurements into structural information still requires assumptions about how the experimental signal enters the folding model [24, 25].

These limitations do not make the calculations uninformative, but they define what can reasonably be concluded from them. Structural analysis can help decide which variants are worth testing, identify assumptions that would otherwise remain implicit, and make an experimental question more specific. The experimental result remains the final test.

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